

Henry Willems

Curriculum Vitae

ul. Rogalińska 1/11
01-206 Warsaw
Poland

* 27 December 1999

+49 157 5721 8840

✉ henry.willems@ncbj.gov.pl

in [henry-willems-ab54b9295](#)

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Education

- 2025/10–
present **PhD in Astrophysics**, *National Centre for Nuclear Research (NCBJ)*, Warsaw, Poland, EQF Level 8
Low Surface Brightness Galaxies in the LSST Era; detection and classification; using Python packages `pyhalomodel`, `halomod` for modelling galaxy power spectra
- 2023/10–
2025/03 **Master of Science in Astrophysics**, *University of Potsdam*, Potsdam, Germany, Grade: 1.5, EQF Level 7
Courses: Astrophysics I & II, Advanced Astrophysics, Methods of Modern Astrophysics, Introductory project and research training.
Thesis: *Determining the physical parameters of eclipsing binaries in the B-Type characterization programme (BBC) with PHOEBE.*
- 2022/10–
2023/09 **Master of Science in Physics**, *University of Potsdam*, Potsdam, Germany, EQF Level 7
Focus: Quantum mechanics, solid state physics, astrophysics (stars and stellar evolution & galaxies and cosmology).
- 2019/10–
2022/10 **Bachelor of Science in Physics**, *University of Potsdam*, Potsdam, Germany, Grade: 1.3, EQF Level 6
Courses: Mathematics for Physicists I-IV, Experimental Physics I-V, Theoretical Physics I-V, Lab courses, Climate Physics, Astronomy.
Thesis: *Hunting for the progenitors of helium white dwarfs.*
Credits: 180

Experience

Vocational

- 04/2025–
07/2025 **Promotional Stipend**, *Max-Planck-Institute for Astronomy*, Heidelberg, Germany
○ Downloading FEROS spectroscopic data from the ESO portal.
○ Organizing and reducing the data (part-time).
- 04/2024–
03/2025 **University Research Assistant**, *Max-Planck Institute for Astronomy*, Heidelberg, Germany
○ Downloading FEROS spectroscopic data from the ESO portal.
○ Organizing and reducing the data (17 hours/week).

- 12/2023- **Cashier, Mittendrin**, Potsdam, Germany
02/2024 ○ Registration of incoming donations.
○ Registration of other incomes and expenses for the community (8 hours/week).

Languages

German	Native speaker	
English	C1	Proficient
French	B1–B2	Independent user

Social Engagement

- 2021/10- **Buddy Program**
2022/03 ○ Helping international students to start their academic lives in Germany, helping with administrative tasks and social life

Skills

MS Office	Word, Excel, PowerPoint, Zoom
Programming	Python (advanced), C++ (basics)
Operating Systems	Linux (Bash/Shell)

Conferences and Workshops

- 2025/07 **Machine Learning in Astrophysical Studies**, Heidelberg, Germany. Learned about new ML applications. ML4ASTRO Workshop 2025.
- 2025/07 **Binary Stars in the Space Era**, Keele, UK. Poster on finished master's thesis. Space Binary 2025.
- 2025/06 **jAG Summer School 2025**, Potsdam, Germany (hosted at the Leibniz Institut für Astrophysik Potsdam). Talk on finished master's thesis. jAG Summer School 2025.
- 2024/09 **Binary and Multiple Stars in the Era of Big Sky Surveys**, Litomyšl, Czech Republic. Talk on ongoing master's thesis. kopal2024.physics.muni.cz.
- 2023/08 **Workshop on Observational Techniques**, Ondřejov, Czech Republic. Telescope imaging, data reduction, final presentations. Workshop 2023.
- 2022/09 **Yearly Meeting of the German Astronomical Society**, Bremen, Germany. Gave talk on bachelor's thesis in hot stars splinter session. AG 2022.

Publications

- 2025/04 H. Willems, *Determining the physical parameters of eclipsing binaries from the B-Type Binaries Characterization Programme (BBC) with PHOEBE, Contributions of the Astronomical Observatory Skalnaté Pleso*, **55**(3), 444–451 (2025). DOI: 10.31577/CAOSP.2025.55.3.444.